

1. Create Your Personal Workspace

Create a directory named StudentLab. Inside it, create the following subdirectories:

- Programs
- Assignments
- Notes

Verify the directory structure and display the current working directory.

```
[jeetpitale@Jeet's StudentLab % mkdir Assignments
[jeetpitale@Jeet's StudentLab % mkdir Programs
[jeetpitale@Jeet's StudentLab % mkdir Notes
[jeetpitale@Jeet's StudentLab % ls
Assignments      Notes            Programs
[jeetpitale@Jeet's StudentLab % pwd
/Users/jeetpitale/DA/Sem1/ITP/Assignment1/StudentLab
[jeetpitale@Jeet's StudentLab % █
```

2. File Management Practice

Inside the Assignments directory:

- Create the files:
 - assignment1.txt
 - assignment2.txt
 - assignment3.txt
- Do all using Command Lines learned in this Lab.
- Rename assignment3.txt to lab3.txt.
- Delete assignment2.txt.
- Display the remaining files.

```
[jeetpitale@Jeet's StudentLab % cd Assignments
[jeetpitale@Jeet's Assignments % cat > assignment1.txt
^C
[jeetpitale@Jeet's Assignments % ls
assignment1.txt
[jeetpitale@Jeet's Assignments % nano assignment2.txt
[jeetpitale@Jeet's Assignments % nano assignment3.txt
[jeetpitale@Jeet's Assignments % ls
assignment1.txt
[jeetpitale@Jeet's Assignments % touch assignment2.txt
[jeetpitale@Jeet's Assignments % touch assignment3.txt
[jeetpitale@Jeet's Assignments % ls
assignment1.txt assignment2.txt assignment3.txt
[jeetpitale@Jeet's Assignments % rm assignment3.txt to lab3.txt
rm: to: No such file or directory
rm: lab3.txt: No such file or directory
[jeetpitale@Jeet's Assignments % ren assignment3.txt to lab3.txt
zsh: command not found: ren
[jeetpitale@Jeet's Assignments % mv assignment3.txt lab3.txt
mv: rename assignment3.txt to lab3.txt: No such file or directory
[jeetpitale@Jeet's Assignments % ls
assignment1.txt assignment2.txt
[jeetpitale@Jeet's Assignments % touch assignment3.txt
[jeetpitale@Jeet's Assignments % ls
assignment1.txt assignment2.txt assignment3.txt
[jeetpitale@Jeet's Assignments % mv assignment3.txt lab3.txt
[jeetpitale@Jeet's Assignments % ls
assignment1.txt assignment2.txt lab3.txt
[jeetpitale@Jeet's Assignments % rm assignment2.txt
[jeetpitale@Jeet's Assignments % ls
assignment1.txt lab3.txt
[jeetpitale@Jeet's Assignments % █
```

3. Creating and Editing Files

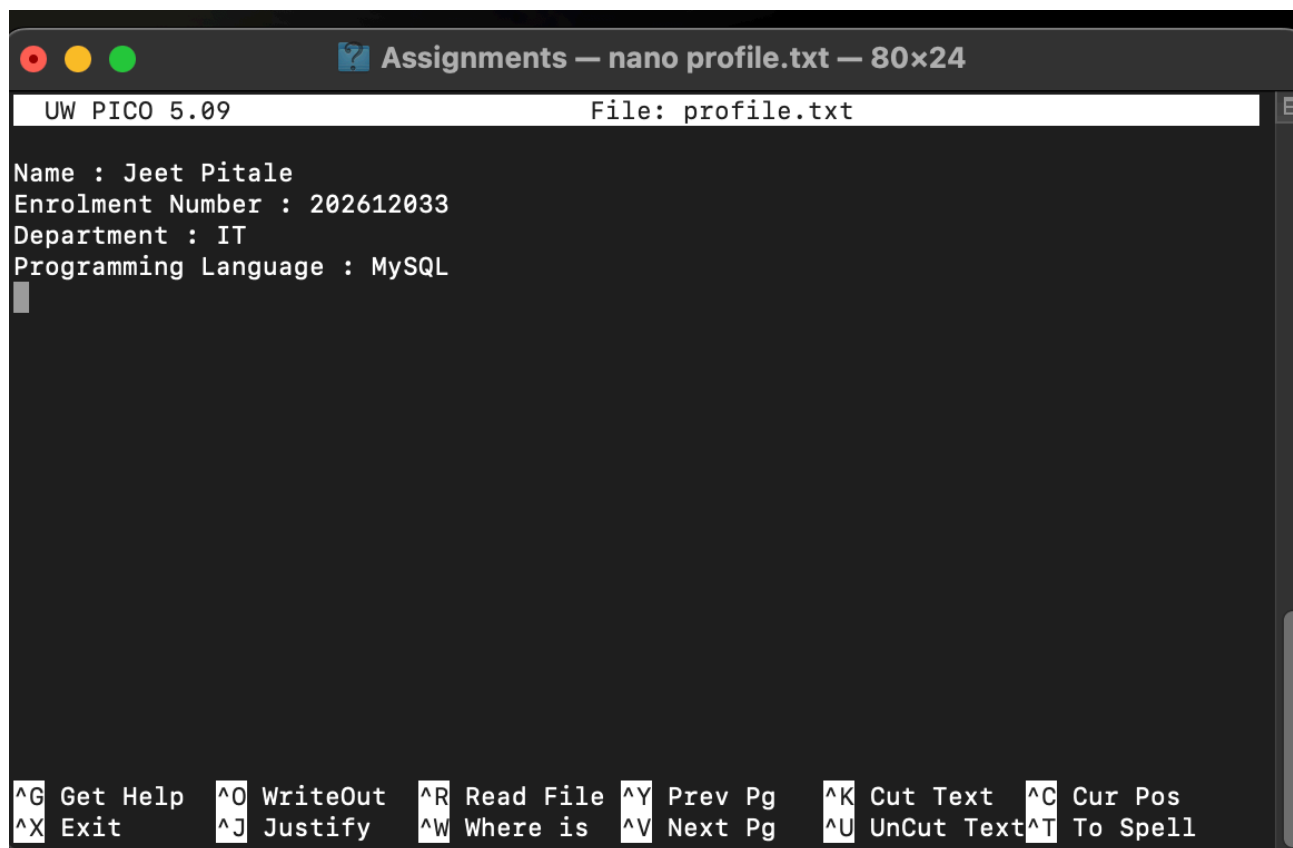
Create a file named profile.txt using the Nano editor.

Enter the following information:

- Name
- Enrollment Number
- Department
- Favourite Programming Language

Save the file and display its contents using the terminal.

```
jeetpitale@Jeet's Assignments % ls
assignment1.txt lab3.txt      profile.txt
jeetpitale@Jeet's Assignments % nano profile.txt
jeetpitale@Jeet's Assignments %
```



The screenshot shows a terminal window titled "Assignments — nano profile.txt — 80x24". The window contains the following text:

```
UW PICO 5.09 File: profile.txt

Name : Jeet Pitale
Enrolment Number : 202612033
Department : IT
Programming Language : MySQL
```

At the bottom of the window, there is a status bar with the following text:

```
^G Get Help ^O WriteOut ^R Read File ^Y Prev Pg ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where is ^V Next Pg ^U UnCut Text ^T To Spell
```

4. File Copy and Backup

Create a file named report.txt.

Copy it to a new file named report backup.txt.

Edit only the backup file and add one additional line.

Finally, display both files and verify that the original file remains unchanged.

```
[jeetpitale@Jeet's Assignments % touch report.txt
[jeetpitale@Jeet's Assignments % nano report.txt
[jeetpitale@Jeet's Assignments % cp report.txt "report backup.txt"
[jeetpitale@Jeet's Assignments % nano "report backup.txt"
[jeetpitale@Jeet's Assignments % cat report.txt
This is the original report.
[jeetpitale@Jeet's Assignments % cat "report backup.txt"

This is the original report.
This line was added to the backup.
jeetpitale@Jeet's Assignments %
```

5. Directory Tree Creation

Create the following directory structure:

LinuxLab

C

C++

Python

Java

Inside each directory, create a file named program1.txt.

Display the contents of every directory.

```
jeetpitale@Jeet's LinuxLab % cd C
[jeetpitale@Jeet's C % ls
program1.txt
jeetpitale@Jeet's C % cd ..
[jeetpitale@Jeet's LinuxLab % cd C++
[jeetpitale@Jeet's C++ % ls
program1.txt
jeetpitale@Jeet's C++ % cd ..
[jeetpitale@Jeet's LinuxLab % cd Python
[jeetpitale@Jeet's Python % ls
program1.txt
jeetpitale@Jeet's Python % cd ..
[jeetpitale@Jeet's LinuxLab % cd Java
[jeetpitale@Jeet's Java % ls
program1.txt
jeetpitale@Jeet's Java %
```

Github Link : <https://github.com/JeetPitaleXdau>